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$$\begin{aligned}\int \frac{x^4 dx}{\sqrt{1-x^{10}}} &= \int \frac{x^4 dx}{\sqrt{1-x^{5 \cdot 2}}} \\ u = x^5 &\Rightarrow du = 5x^4 dx \Rightarrow x^4 dx = \frac{1}{5} du \\ \int \frac{x^4 dx}{\sqrt{1-x^{10}}} &= \int \frac{1}{\sqrt{1-u^2}} \cdot \frac{1}{5} du = \frac{1}{5} \int \frac{1}{\sqrt{1-u^2}} du \\ &= \frac{1}{5} \operatorname{Bgsin}(u) + c = \frac{1}{5} \operatorname{Bgsin}(x^5) + c\end{aligned}$$

References